

QualiAdept: IT Masterclass

- How technology really works



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Module 1: Hardware - Building "Thinking" from Matter

Session 1: The Processor (CPU) - Architecture and Execution (7 Chapters)

- **Chapter 1:** The Transistor - The Switch That Changed the World.
- **Chapter 2:** Von Neumann Architecture - The foundation of any modern computer.
- **Chapter 3:** Internal Units: ALU (Arithmetic Unit) and CU (Control Unit).
- **Chapter 4:** Registers - Ultra-fast memory next to the "heart".
- **Chapter 5:** The Instruction Cycle (Fetch-Decode-Execute) - In detail.
- **Chapter 6:** Frequency, Cores and Threads.
- **Chapter 7:** Manufacturing process (Nanometers) and Moore's Law.

Session 2: Memory Hierarchy and Buses (6 Chapters)

- **Chapter 1:**RAM - From capacitors to bits (DRAM vs. SRAM).
- **Chapter 2:**Cache Memory (L1, L2, L3) - The bridge over the speed bottleneck.
- **Chapter 3:**Permanent Storage: HDD vs. SSD (SATA, NVMe, NAND Flash).
- **Chapter 4:**Virtual Memory and Paging - How to expand RAM to disk.
- **Chapter 5:**ROM and BIOS/UEFI - The first breath of life of the machine.
- **Chapter 6:**Motherboard and Chipset - Data highways (Buses, PCIe, USB).

Module 2: Digital Logic and Basic Software

Session 3: The Binary Universe and Logic Gates (6 Chapters)

- **Chapter 1:**Binary and Hexadecimal System - How machines count.
- **Chapter 2:**Text Representation: ASCII, UTF-8 and Unicode.
- **Chapter 3:**Representation of Images and Sound (Pixels, Colors, Sampling).
- **Chapter 4:**Basic Logic Gates (AND, OR, NOT, NAND, NOR, XOR).
- **Chapter 5:**Adders and Logic Circuits - How to physically add 1 + 1.
- **Chapter 6:**Boolean Algebra - The mathematics behind digital decisions.

Session 4: The Operating System - Chaos Management (6 Chapters)

- **Chapter 1:**What is an Operating System? The role of mediator between humans and hardware.
- **Chapter 2:**The Kernel - The Invisible Heart (User Space vs. Kernel Space).
- **Chapter 3:**Process Management - How the OS divides CPU time (Scheduling & Context Switching).
- **Chapter 4:**Memory Management - Allocation, protection and Garbage Collection.
- **Chapter 5:**File System (NTFS, ext4, APFS) - How we organize bits on the disk.
- **Chapter 6:**Drivers and Interrupts - How the OS communicates with peripherals.

Module 3: From Code to Image(Software & Graphics)

Session 5: Programming Languages, Compilation and

Execution (6 Chapters)

- **Chapter 1:** Levels of Abstraction - From Assembly to High-Level Languages.
- **Chapter 2:** Compiler vs. Interpreter - Translation into machine code.
- **Chapter 3:** Virtual Machines and Hybrid Languages (JVM, CLR, JIT Compilation).
- **Chapter 4:** Anatomy of a Running Program: Stack and Heap.
- **Chapter 5:** Classic Runtime Errors: Stack Overflow, Memory Leaks and Null Pointers.
- **Chapter 6:** Software Life Cycle (From source code to distribution).

Session 6: Graphics, Peripherals and Interface (5 Chapters)

- **Chapter 1:** GPU vs. CPU - Why do we need massively parallel computing?
- **Chapter 2:** Anatomy of a Screen Image: Pixels, Framebuffer and Refresh Rate.
- **Chapter 3:** 3D Rendering, Shaders and Ray Tracing - How we build virtual worlds.
- **Chapter 4:** Input Peripherals - Convert physical actions into digital signals.
- **Chapter 5:** Digital Audio - From sound wave to analog signal and PCM.

Module 4: The Interconnected World and Modern Technologies

Session 7: Networks, Internet and Protocols (7 Chapters)

- **Chapter 1:** What is a Network? The OSI Model and the TCP/IP Stack explained simply.
- **Chapter 2:** IP Addresses (IPv4 vs. IPv6) and MAC Address - Identity on the Network.
- **Chapter 3:** DNS (Domain Name System) - The phone book of the Internet.
- **Chapter 4:** The Data Road: Packets, Routers, Switches and Submarine Cables.
- **Chapter 5:** Web protocols: HTTP/HTTPS and TCP vs. UDP.
- **Chapter 6:** Network Security: Firewall, VPN and DDoS Attacks.
- **Chapter 7:** Wireless Transmission: Wi-Fi, 5G and Fiber Optics.

Session 8: Cloud, Cryptography and Artificial Intelligence (6 Chapters)

- **Chapter 1:** What is the Cloud? (IaaS, PaaS, SaaS and Virtualization/Containers).
- **Chapter 2:** Cryptography - How we protect data (Symmetric vs. Asymmetric Encryption /

RSA).

- **Chapter 3:**SSL/TLS Certificates and Digital Signature.
- **Chapter 4:**Databases (SQL vs. NoSQL) - Where the world stores big data.
- **Chapter 5:**Fundamentals of Artificial Intelligence - Neural Networks and Data Modeling.
- **Chapter 6:**The Big Picture - What happens in the millisecond you press Enter in a browser.